

Constructing and Solving Equations — Paper D

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every equation on this paper is designed to come out exactly.
- The method is not printed on this paper. You are expected to know it.
- When a bracket is subtracted, the minus sign multiplies every term inside it. When a fraction wraps a whole side, multiplying by the denominator undoes it in one move.
- Several questions ask you to explain, prove or diagnose. A correct calculation is not by itself an explanation — say what the algebra shows.
- The mark for each question is shown in square brackets on the right.
- If a question stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – BRACKETS AND FRACTIONS

1. Solve:

(a) $6(x - 2) = 4(x + 1)$ [2]

(b) $3(2x + 5) - 4(x - 1) = 27$ [2]

2. Solve:

(a) $(3x - 2) \div 4 = 4$ [2]

(b) $x \div 2 + x \div 3 = 10$

[2]

SECTION 2 – EQUATIONS THAT DESCRIBE SOMETHING

3. A rectangle has length $(3x - 1)$ cm and width $(x + 2)$ cm. Its perimeter is 42 cm.

(a) Form an equation in x and simplify it.

[2]

(b) Solve it, then write down the length and the width and check the perimeter.

[2]

4. Two plumbers quote for a job.

Plumber A: a 250 rupee call-out fee, then 180 rupees an hour.

Plumber B: no call-out fee, but 230 rupees an hour.

(a) Write an expression for each plumber's charge for a job lasting h hours.

[1]

(b) Form and solve an equation to find the length of job for which they charge the same.

[2]

(c) Who is cheaper for a 3-hour job, and by how much?

[1]

5. Four consecutive whole numbers add up to 90.

- (a)** Using n for the smallest, form and solve an equation, and write down the four numbers. [2]

- (b)** By factorising the expression for the sum, show that the sum of four consecutive whole numbers is always even. [2]

SECTION 3 – MARKING SOMEBODY ELSE

6. Two students were each asked to solve $5x - 3 = 2x + 12$. [5]

Student T wrote: " $5x + 2x = 12 + 3$, so $7x = 15$, so $x = 15/7$ "

Student U wrote: " $3x = 9$, so $x = 3$ "

For each student, describe exactly what they did wrong. Then give the correct solution.

7. A student has written two solutions in his book, and both are wrong. [5]
- (i) " $3x + 6 = 18$, so dividing by 3 gives $x + 6 = 6$, so $x = 0$."
- (ii) " $4(x + 3) = 20$, so $4x + 3 = 20$, so $4x = 17$, so $x = 4.25$."
- For each one, explain the mistake and give the correct answer. Then state the single check that would have caught both of them.

End of paper. Check every answer has a unit or a form the question actually asked for.